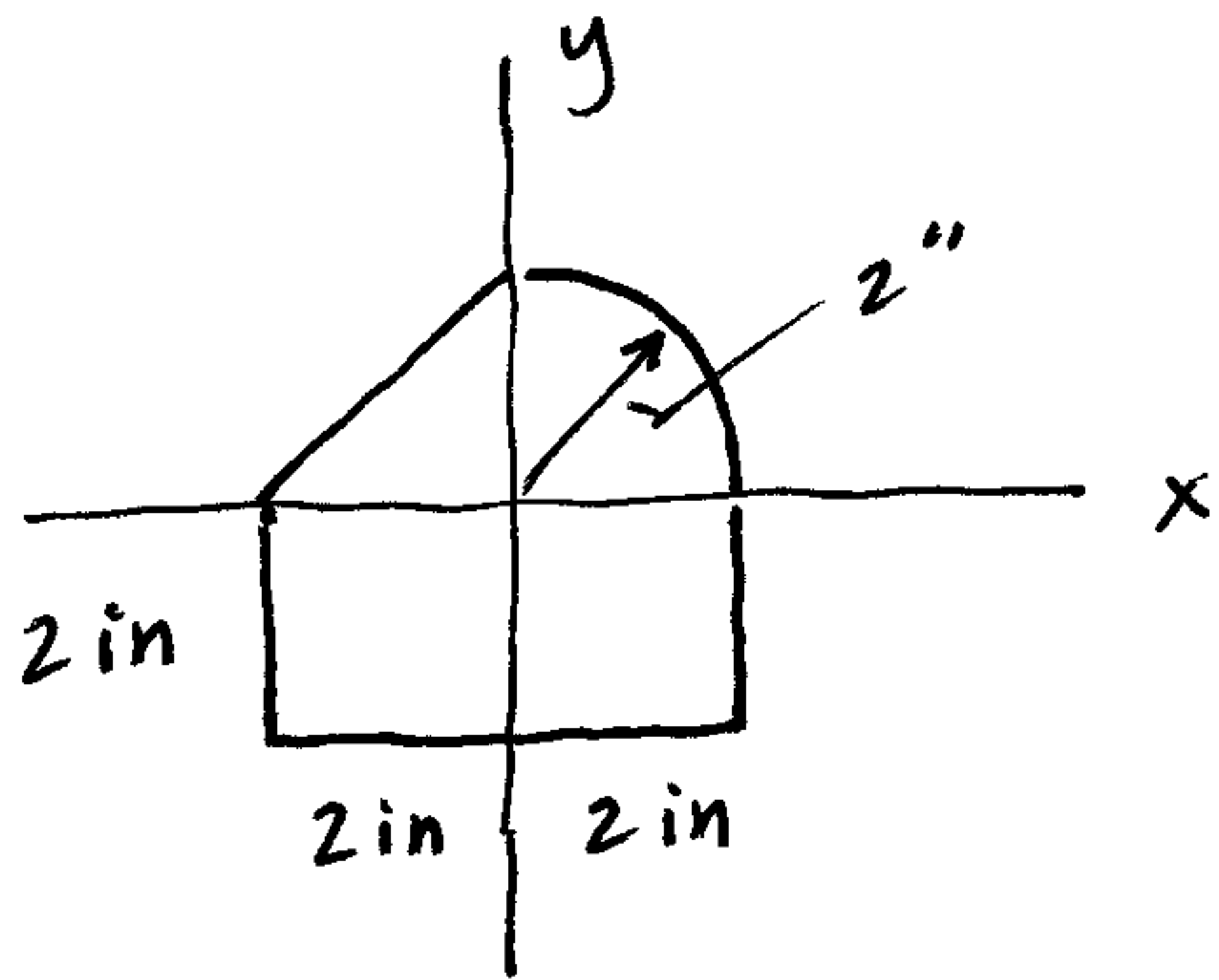


Composite Area MOI Ex Prob 4



Determine I_x, I_y

All shapes touch axes...



$$I_x = \left[\frac{1}{3} (4)(2)^3 \right]_{\square} + \left[\frac{1}{12} (2)(2)^3 \right]_{\Delta} + \left[\frac{\pi}{16} (2)^4 \right]_{\Delta}$$

$$= 10.67 + 1.333 + \pi$$

$$\boxed{I_x = 15.1 \text{ in}^4}$$

$$I_y = \left[\frac{1}{12} (2)(4)^3 \right]_{\square} + 1.333 + \pi$$

$$= 10.67 + 1.333 + \pi$$

$$\boxed{I_y = 15.1 \text{ in}^4}$$

Why same?

$$I_x = \triangle + \Delta + \square + \square$$

$$I_y = \triangle + \Delta + \square + \square$$

I_{xy} (Use I_{ORIGIN} Equations)!

$$I_{xy} = \left[-.04167(2)^2(2)^2 \right]_{\Delta} + \left[\frac{1}{8} (2)^4 \right]_{\Delta} + \left[0 + 2(4)(0)(-1) \right]_{\square}$$

$$I_{xy} = -.667 + 2 + 0$$

$$\boxed{I_{xy} = 1.33 \text{ in}^4}$$

USE
 I_{ORIGIN}
EQNS